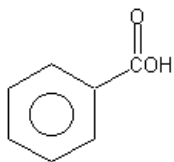


Organic- Answer Section MULTIPLE CHOICE

1 B	2 C	3 D	4 A	5 D	6 D	7 D
8 C	9 C	10 C	11 A	12 A	13 E	14 B
15 C	16 D	17 C	18 B	19 C	20 A	

SHORT ANSWER

21. ANS:

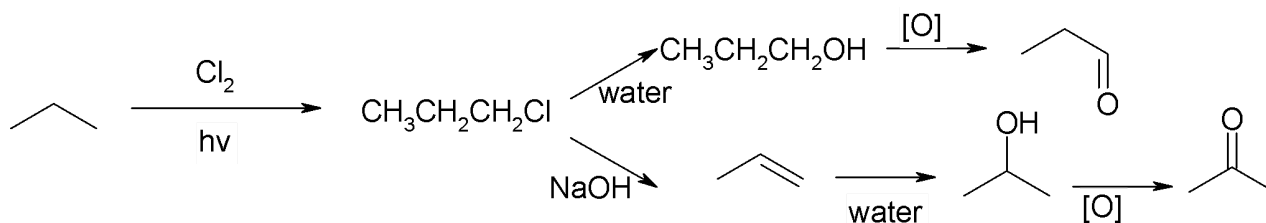


- derivatives & breakdown products in the body could include benzene and other aromatics known to be carcinogens.
- preservatives and other organic compounds may cause unforeseen health side effects (long or short term) when ingested if they are not normally a chemical found in our bodies
- high levels increase the risks of side effects
- allergies, illness and even links to cancer are possible
- some organic compounds react with the body's hormones or mimic the behaviour of natural hormones, and side effects result

22. ANS:

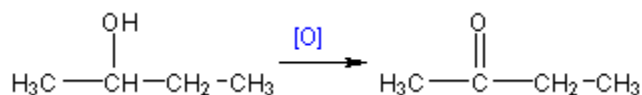
- $(C_2H_3Cl)_n + 5/2 n O_2 \rightarrow 2nCO_2 + nH_2O + nHCl$
- HCl produced, is a strong acid
- choking is caused by its corrosive influence
- other nitrogen containing products can also provide cyanide compounds ($-C\equiv N:$) during a fire.

23. ANS:



'Tis true -- the hardest question on the test -- Mr. S

24. ANS:

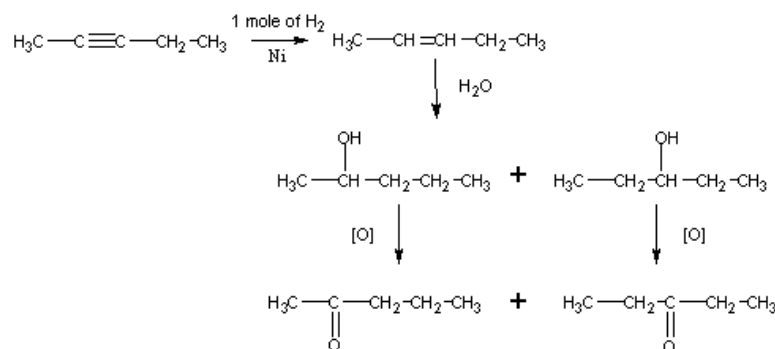


- add water and diethylether
- 2-butanol would dissolve in the water (polar) and 2-butanone would dissolve in the diethylether
- alternate, if both are liquids, if boiling point are distinct enough distillation

25. ANS:

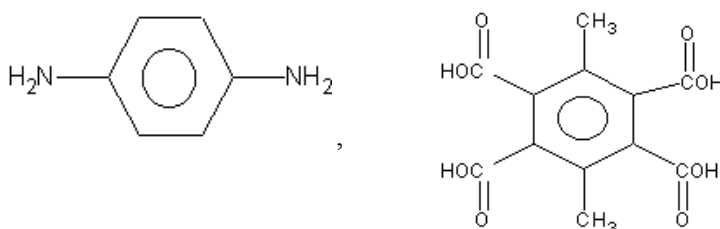
- aromatic ring
- two amide groups
- carboxylic acid group
- (thioether group, -S-)

26. ANS:

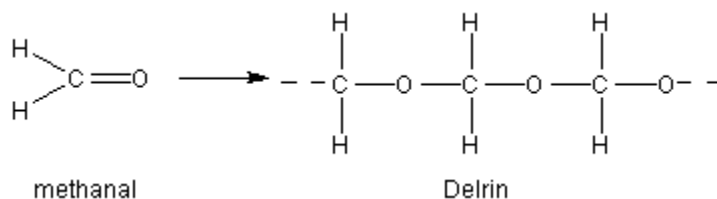


27. ANS:

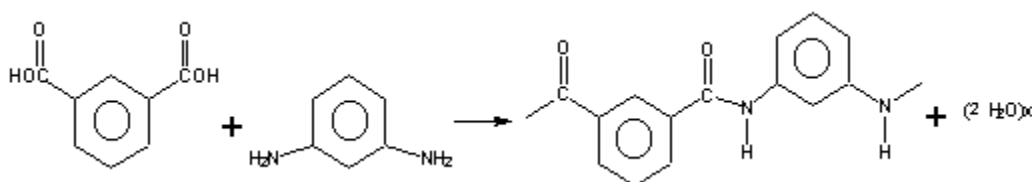
- is a polyamide polymer
- monomers are:



28. ANS:



29. ANS:



condensation reaction

30. ANS: (and ideally the $-\text{COOH}$ and $-\text{NH}_2$ ends should be indicated) P.S. which combo is missing?

Each of the ones below could also be re-written in the opposite order

- | | |
|-----------|---------------|
| - gly-ala | - gly-ala-ser |
| - gly-ser | - gly-ser-ala |
| - ala-ser | - ala-gly-ser |

ESSAY

31. ANS: answers will vary

- during a synthesis, minor products are often created in addition to major ones
- this will require constant purification since minor products could be very harmful
- isomers are often some of the minor products and are difficult to separate since they have physical and chemical properties similar to desired products
- chiral isomers can be produced and be detrimental in biological systems where chirality is often important to enzymes
- administering the drug: the digestion system or the pH of the blood may destroy it – IT IS A CHALLENGE TO GET IT INTO THE CELL or INTO THE NUCLEUS IF NEEDED.
- unforeseen reactions with hormones, enzymes or other cellular proteins can cause side effects from other products

32. ANS: answers may vary

- cellulose: occurs in plants as a polymer of glucose
- since it is insoluble in water and strong, it is used as a building material (i.e., wood), often pressure-treated to increase its life span; paper products are also made from cellulose
- silk is spun by the silk worm: it is dyed and used as a fabric
- cotton is produced by the cotton plant: it is dyed and used as a fabric
- rubber is from the sap of a rubber tree: it is vulcanized to increase stiffness by cross-linking
- DNA in our cells: genetic scientists study the blue prints of life; biotechnologists manipulate DNA in other organisms to create medicines, hormones, etc.
- protein in living organisms: replacement proteins are synthesized to cure diseases, help nutrition
- keratin in nails and hair: cosmetic changes in hair with perms, etc.
- amber is a natural resin: fossil records, jewelry
- wool from sheep: dyed and made into fabrics
- glycogen is a glucose polymer in humans, for storage of energy
- starch is a glucose polymer found in many plants: provides a source of energy for the human body